

Clean Water and Sanitation Assistant Chatbot

1. Selected SDG and Reason for Selection

Selected SDG

SDG 6 – Clean Water and Sanitation

What the SDG Focuses On

SDG 6 focuses on ensuring availability and sustainable management of water and sanitation for all. It promotes safe drinking water, proper sanitation facilities, wastewater management, and awareness about water conservation.

Why This Issue Is Important

Clean water is essential for human health, agriculture, and sustainable development. Water pollution, shortage of clean drinking water, and poor sanitation facilities create serious health and environmental problems. Awareness about water conservation and hygiene practices can help communities protect water resources.

2. Problem Statement

Many people lack awareness about water conservation, water pollution prevention, and proper sanitation practices. Water wastage, contamination of water sources, and improper waste disposal continue to affect communities.

Where Does This Problem Exist?

This problem exists in homes, schools, villages, cities, industries, and public areas.

Who Is Affected?

- Students
- Families
- Communities
- Farmers
- Future generations

Why Is It Serious?

Unsafe water and poor sanitation can cause health problems, environmental pollution, and shortage of water resources.

What Could Happen If It Is Not Solved?

Water scarcity, increased pollution, and health risks will continue to grow, affecting quality of life and sustainable development.

3. Project Objective

The objective of this project is to develop an AI-powered chatbot that educates users about clean water management and sanitation practices. The chatbot provides guidance on water conservation, hygiene, pollution prevention, and sustainable water usage.

4. Proposed Solution

Project Name

Clean Water and Sanitation Assistant Chatbot

What Is The Project?

The Clean Water and Sanitation Assistant Chatbot is an AI-based chatbot developed using Botpress. It helps users learn about water conservation, sanitation, hygiene practices, and protection of water resources.

How The System Works

Users interact with the chatbot by asking questions related to clean water, sanitation, water saving methods, and pollution control. The chatbot processes user queries and provides suitable responses.

How It Helps Solve The Problem

The chatbot spreads awareness about clean water practices and encourages people to adopt responsible water usage habits.

Who Will Use The Application?

- Students
- Teachers
- Environmental enthusiasts
- General public

5. Project Features

Clean Water Awareness

Provides information about the importance of safe and clean water.

Water Conservation Tips

Suggests methods to reduce water wastage and save water.

Sanitation Guidance

Educates users about hygiene and sanitation practices.

Water Pollution Prevention

Provides information about preventing contamination of water sources.

Wastewater Management Awareness

Explains the importance of proper wastewater treatment.

Fallback Response

Handles unrelated questions and redirects users toward clean water and sanitation topics.

6. Technology Stack

- Botpress
- Artificial Intelligence (AI)
- Natural Language Processing (NLP)
- GitHub

7. Project Screenshots

Figure 1: Bot Dashboard/Home Page

Shows the chatbot development environment and configuration.

Figure 2: Workflow Diagram

Shows chatbot flow including user input processing and response generation.

Figure 3: Autonomous Node Instructions

Displays chatbot instructions related to water conservation and sanitation.

Figure 4: Chatbot User Interactions

Shows conversations related to clean water, hygiene, and sanitation awareness.

Figure 5: Deployed Chatbot Interface

Shows the publicly available chatbot interface.

8. Open Source Repository and Project Link

GitHub Repository

(Add your GitHub repository link)

Project Demo Link

<https://cdn.botpress.cloud/webchat/v3.6/shareable.html?configUrl=https://files.bpcontent.cloud/2026/06/22/16/20260622163533-328MHIXB.json>

9. Future Scope

- Voice-enabled chatbot interaction
- Mobile application integration
- Multi-language support
- Real-time water quality updates
- Personalized water-saving recommendations
- Integration with smart water monitoring systems

10. Conclusion

The Clean Water and Sanitation Assistant Chatbot was developed to support SDG 6: Clean Water and Sanitation. The chatbot provides users with information about water conservation, sanitation, hygiene, and pollution prevention. It demonstrates how Artificial Intelligence can be used to spread awareness and contribute to solving real-world water and sanitation challenges.